



NEXT CHAPTER @ SMITH PUBLIC LIBRARY

Finding Your Way on a Graph

Session 5 — Linear Equations & Graphing Basics

Monday Evenings • 5:30–7:30 PM • Library Conference Room

Tonight's Agenda

1

Welcome & quick recap

10 min

2

The coordinate plane & plotting points

25 min

3

Graphing a line from an equation

30 min

4

Slope & intercept: what they mean

25 min

5

Worked real-world example

15 min

6

Wrap-up & take-home preview

15 min

Quick Recap: Last Week

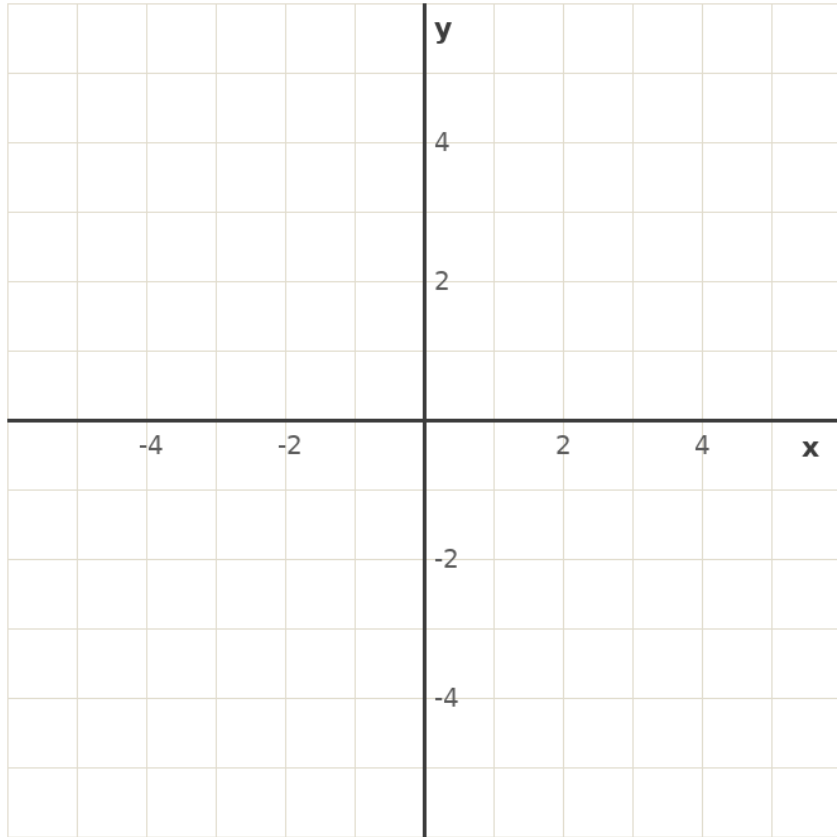
The Balance Rule:

whatever you do to one side of an equation, you do to the other side too.

Tonight:

we take equations like the ones you solved last week and learn to picture them — as lines on a graph.

The Coordinate Plane



Two number lines, crossing at zero.

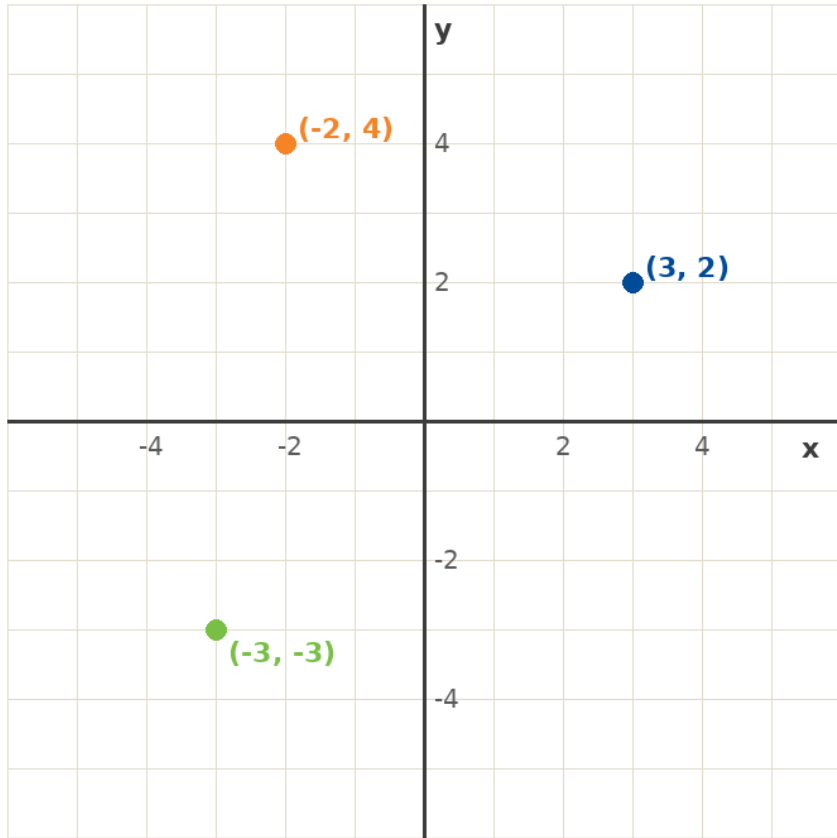
x-axis: the horizontal line (left-right)

y-axis: the vertical line (up-down)

origin: where they cross, (0, 0)

Every point is named by an ordered pair: **(x, y)**

Plotting Points



For a point (x, y) :

1. Move left/right first (the x-value)
2. Then move up/down (the y-value)

Example: $(3, 2)$ means go 3 right, then 2 up.

Negative numbers go the opposite direction — left instead of right, down instead of up.

Your Turn: Coordinates

GUIDED PRACTICE

1

A point is 3 units right and 4 units up from the origin. What are its coordinates?

2

A point is 2 units left and 5 units down from the origin. What are its coordinates?

3

What is the y-coordinate of the point $(6, -3)$?

4

In which quadrant (upper-left, upper-right, lower-left, lower-right) is the point $(-4, 2)$ located?

Let's Check Your Answers



1. $(3, 4)$



2. $(-2, -5)$

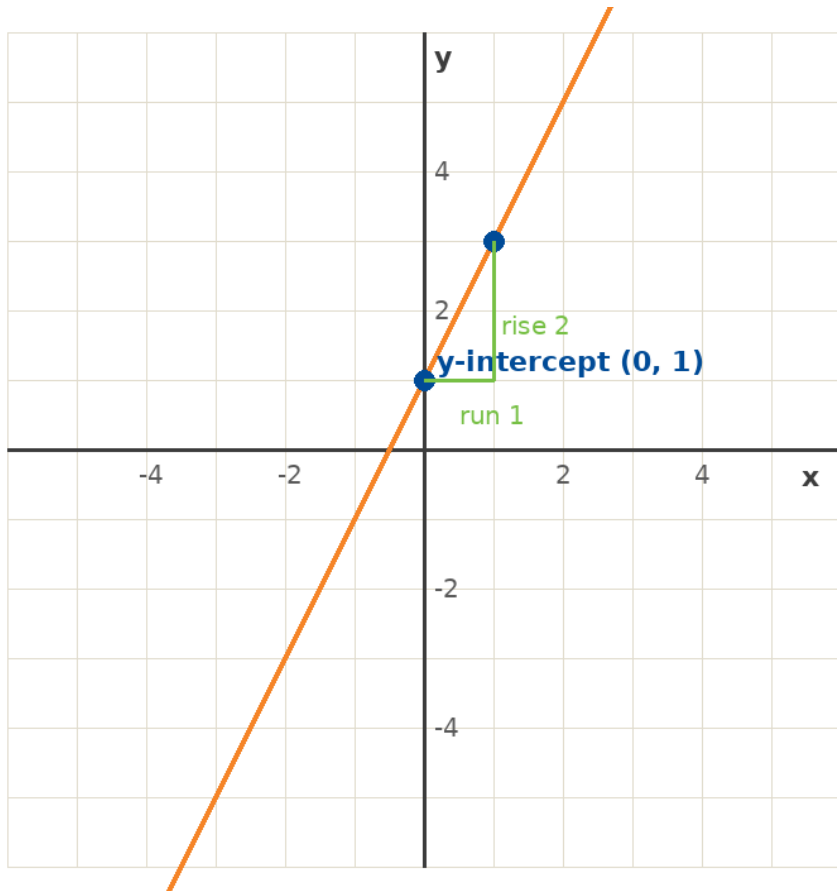


3. -3 (the second number in the pair)



4. Upper-left (negative x , positive y)

Graphing a Line from an Equation



Slope-intercept form: $y = mx + b$

m = the slope (steepness)

b = the y-intercept (where the line crosses the y-axis)

Example: $y = 2x + 1$

Start at the y-intercept (0, 1). Slope 2 means: up 2, right 1 — repeat to find more points.

Your Turn: Points on a Line

GUIDED PRACTICE

1

For $y = x + 3$, what is y when $x = 2$?

2

For $y = 2x - 1$, what is y when $x = 0$? (This also tells you the y -intercept!)

3

What is the y -intercept of $y = 4x + 7$?

4

What is the slope of $y = 5x - 2$?

Let's Check Your Answers



1. $y = 2 + 3 = 5$



2. $y = 2(0) - 1 = -1$ (the y-intercept is -1)



3. 7 (the number added at the end)



4. 5 (the number multiplied by x)

What Slope & Intercept Really Mean



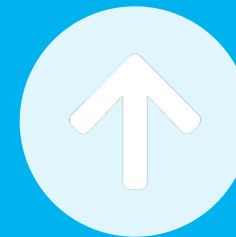
Slope = Steepness

How fast the line rises or falls. A bigger number means a steeper line. Negative means it goes downhill.



Intercept = Starting Point

Where the line begins on the y-axis — the value when x is zero.



Rise Over Run

Slope as a fraction: how much you go up (rise) for how much you go right (run).

Your Turn: Slope & Intercept

GUIDED PRACTICE

1

A line has slope 2 and y-intercept 3. Write its equation in $y = mx + b$ form.

2

What is the slope of the line $y = -3x + 6$?

3

What is the y-intercept of $y = x - 8$?

4

A line passes through $(0, 4)$ with slope 1. Write its equation.

Let's Check Your Answers



1. $y = 2x + 3$



2. -3



3. -8



4. $y = x + 4$

Let's Solve One Together

A phone plan costs \$20 per month, plus \$0.10 per text message. Write an equation, then find the cost for 50 texts.

Set it up: $y = 0.10x + 20$ (x = number of texts, y = total cost)

The slope (0.10): cost per text

The intercept (20): the base monthly fee, even with zero texts

For 50 texts: $y = 0.10(50) + 20 = 5 + 20 = \25

Total cost: \$25

Before You Go



Try tonight's take-home practice

Real practice questions on tonight's material — 10 minutes, any time this week.



Thursday office hours are open

If graphing still feels unfamiliar, bring a specific equation and we'll work through it.



Next week: Geometry & calculator

The final Math session — shapes, measurement, and the on-screen calculator.



You Can Read a Graph.

See you next Monday — geometry and the calculator.

Before next week:

Try tonight's take-home practice. Want more? GED.com's free test preview tutorials (tools.ged.com) are always open, in English or Spanish.